

Constructive Distortion: Improving MLLMs with Attention-Guided Image Warping



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VLMs err when they shouldn't

What device is below the artwork?



LLaVA: *Speaker* ❌

Liu, Haotian, Chunyuan Li, Yuheng Li, and Yong Jae Lee. "Improved baselines with visual instruction tuning." In CVPR 2024

Is there a bottle in the image?

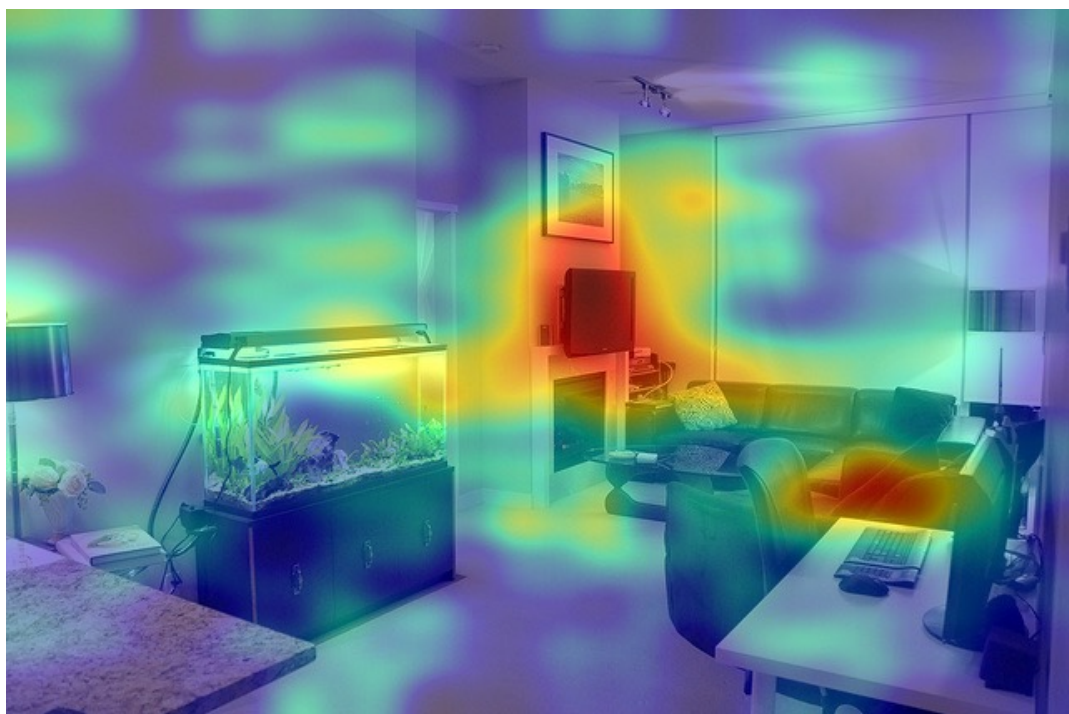


Qwen: No ❌

Bai, Shuai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibong Song, Kai Dang et al. "Qwen2.5-vl technical report." arXiv preprint arXiv:2502.13923 (2025).

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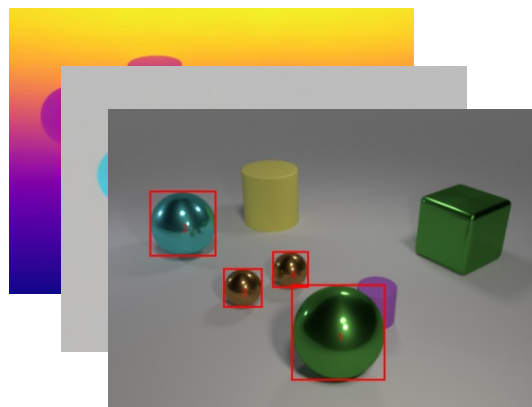
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Past works on improving VLM's for VQA

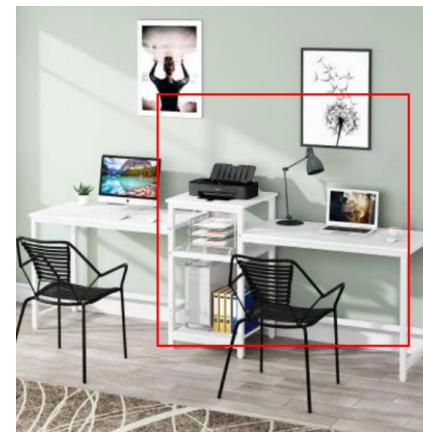


Vision-Language
Decomposition [1]



[cyclist_1] — riding — [bicycle_1]
|
wearing
|
[backpack] — carrying — [dog]

Scene Graph Guided
answering [2]



Cropping [3]



Visual Grounding [4]

[1] Compositional Reasoning via Joint Image and Language Decomposition
Dwip Dalal*, Madhav Kanda*, Zhenhailong Wang, Heng Ji, Unnat Jain
EACL 2026

[2] Compositional Chain-of-Thought Prompting for Large Multimodal Models
Chancharik Mitra, Brandon Huang, Trevor Darrell, Roei Herzig
CVPR 2026

[3] Training-free Perception of Small Visual Details with Multimodal LLMs
Jiarui Zhang, Mahyar Khayatkhoei, Prateek Chhikara, Filip Ilievski
ICLR 2025

[4] Set-of-Mark Prompting Unleashes Extraordinary Visual Grounding in GPT-4V
Jianwei Yang, Hao Zhang, Feng Li, Xueyan Zou, Chunyuan Li, Jianfeng Gao
<https://arxiv.org/abs/2310.11441>

“On the right desk, what is to the left of the laptop?”



(Ours)

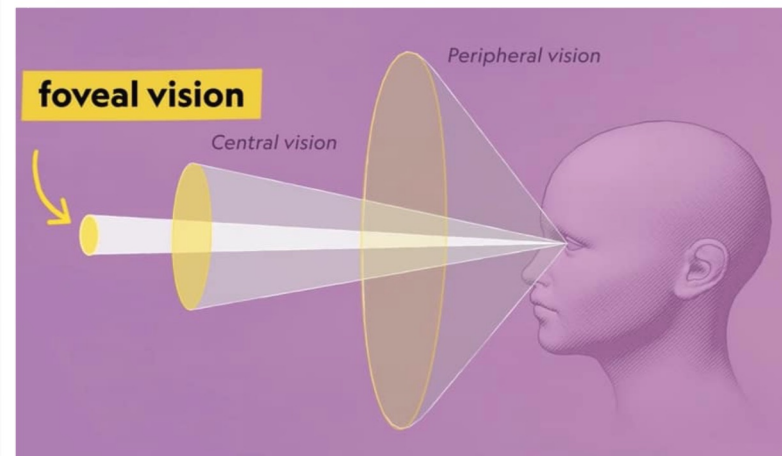
VLM (Image, Query):
"To the left of the laptop there is a **stack of books** on the desk."

VLM (Warped Image, Query):
"There is a **lamp** to the left of the laptop on the right desk."

A new take at foveal vision

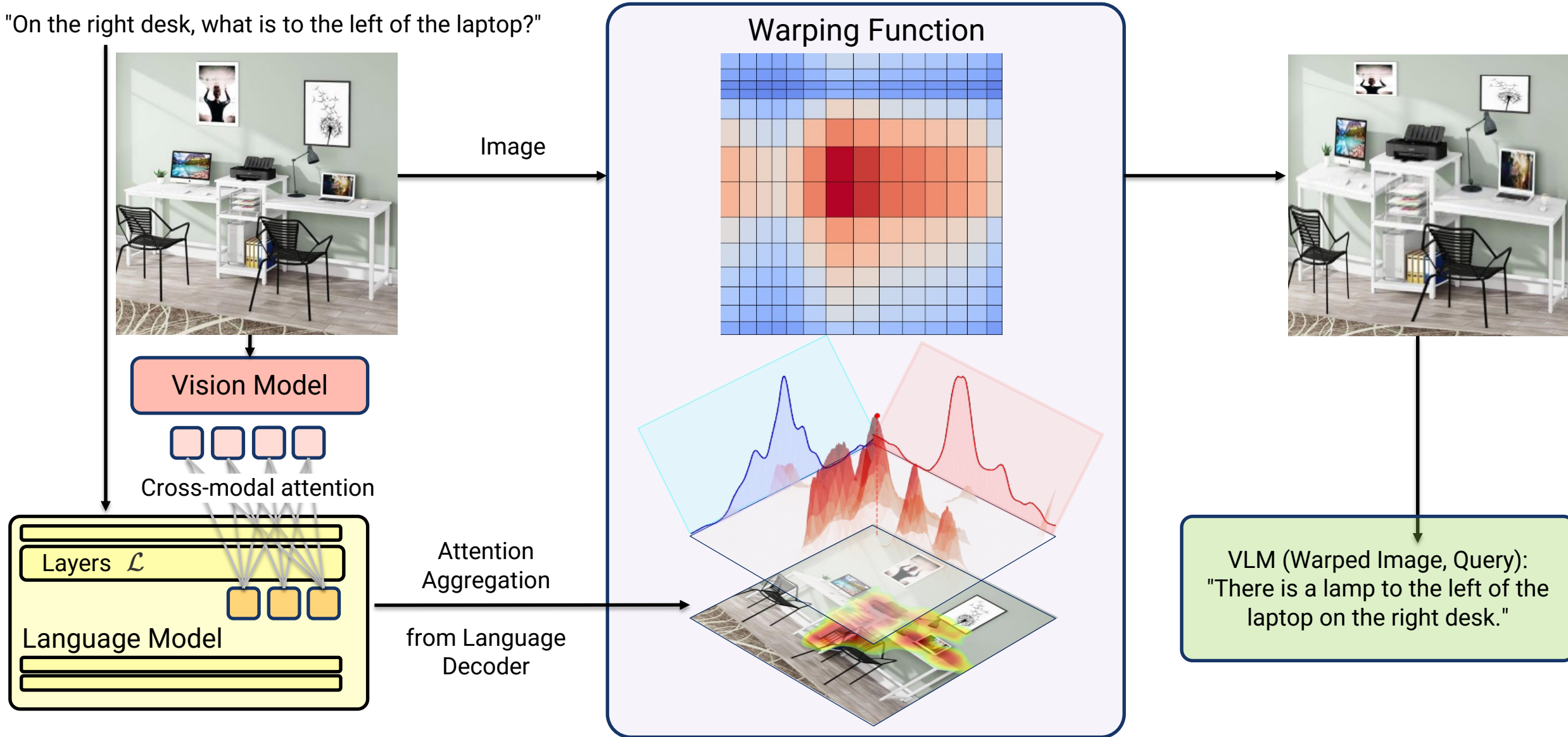


Two fixation points were used. One was the pinhead at the mid-point of the adjustable line—the 'foveal' fixation point; the other was a small cross on the circular screen, which moved with the adjustments of the line, placed at a fixed distance of 4 cm from the mid-point of the line—the 'peripheral' fixation point. The 4 cm distance represented a visual angle of 5.7° . With this arrangement the line could be fixated either by holding the eyes on the pinhead for 5 minutes (foveal fixation), or by holding them on the cross (peripheral fixation). In turn, the test for after-effect could be made with eyes on the pinhead (foveal test) or on the cross (peripheral test). Both foveal fixation with foveal test and peripheral fixation with peripheral test gave a large after-effect, since the area tested was that in which adaptation had occurred.



Attention-Guided Warping (AttWarp)

"On the right desk, what is to the left of the laptop?"

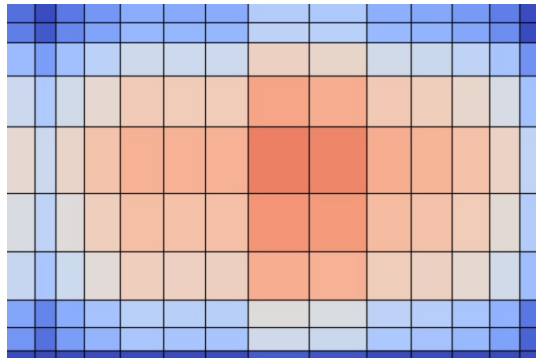


Attention-Guided Warping (AttWarp)

Query: What device is below the artwork?



LLaVA: Speaker



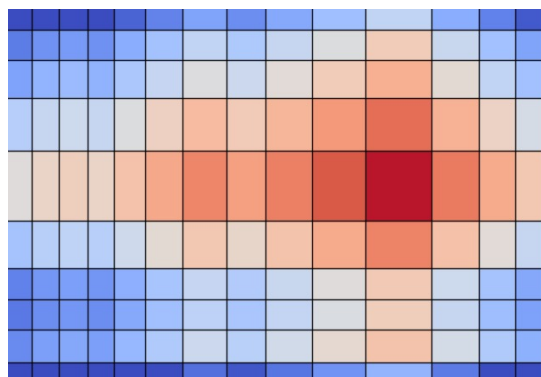
AttWarp: Television



Query: Is there a bottle in the image?



Qwen: No



AttWarp: Yes

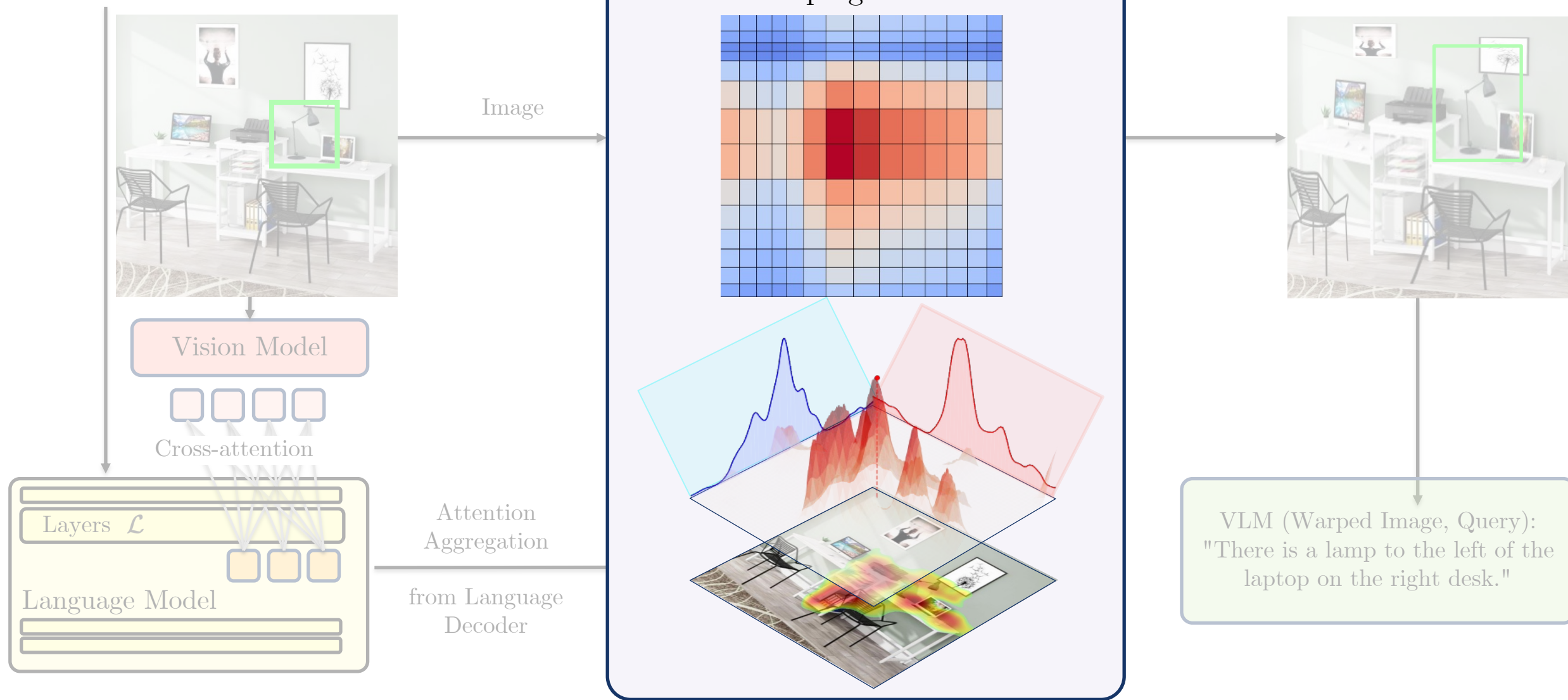


Warping helps across the board!

Methods	Key Technique	TextVQA	GQA	MMMU	POPE	DocVQA
LLaVA (Liu et al., 2024a) (<i>MLP vision-language connector & open data</i>)						
Base MLLM		49.3	60.5	36.9	85.3	18.1
AttWarp (<i>ours</i>)	Rectilinear warping	58.1	63.7	40.4	87.5	25.5
Qwen (Yang et al., 2024a) (<i>Cross-attention VL adapter & partially closed data</i>)						
Base MLLM		81.0	62.4	47.3	86.1	77.3
AttWarp (<i>ours</i>)	Rectilinear warping	84.7	64.0	50.4	87.4	84.1

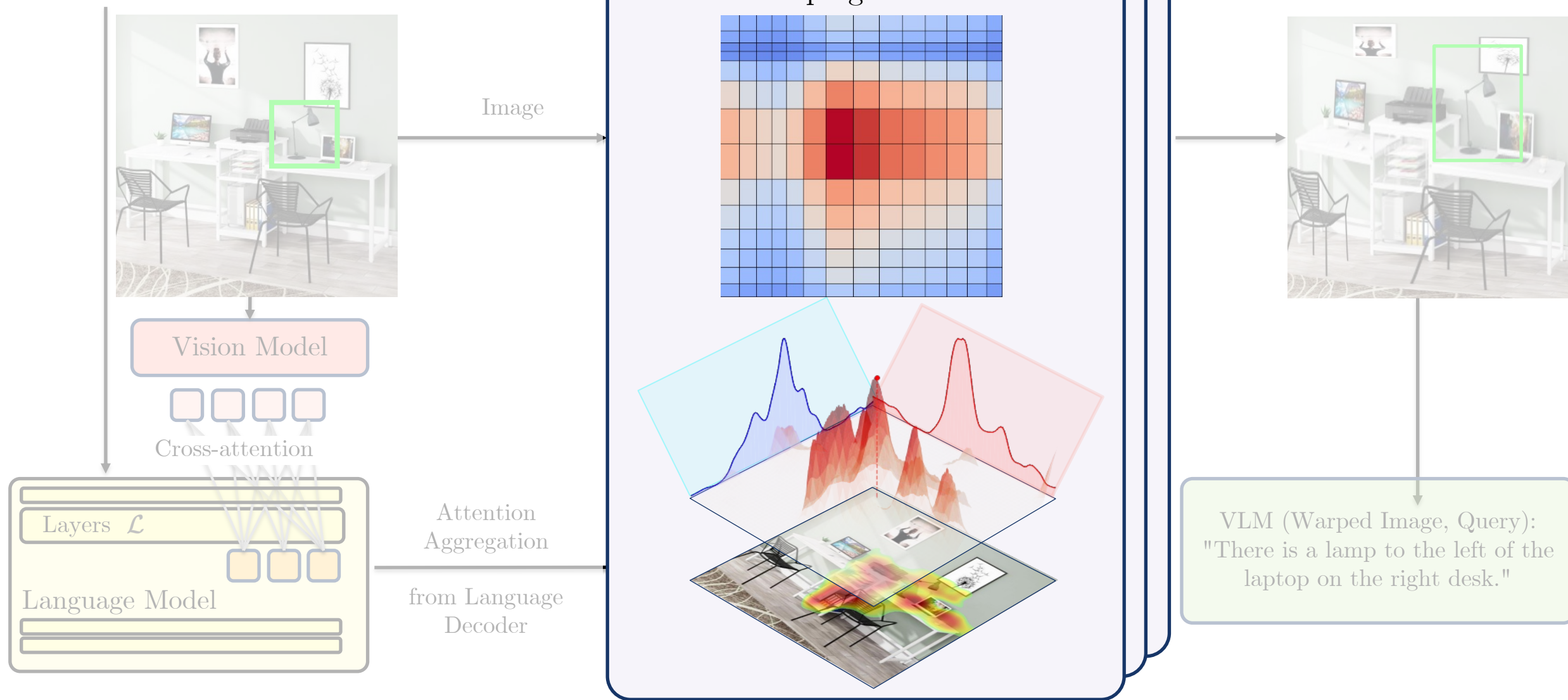
Attention-Guided Warping (AttWarp-Chain)

"On the right desk, what is to the left of the laptop?"



Attention-Guided Warping (AttWarp-Chain)

"On the right desk, what is to the left of the laptop?"



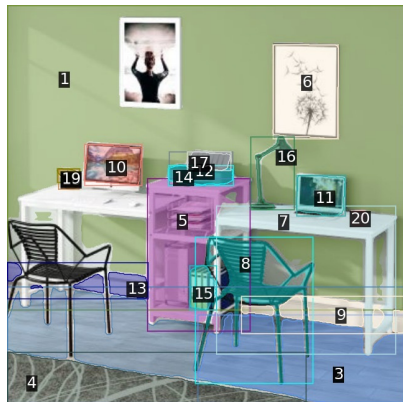
Warping w/ Chaining helps across the board!

Methods	Key Technique	TextVQA	GQA	MMMU	POPE	DocVQA
LLaVA (Liu et al., 2024a) (<i>MLP vision-language connector & open data</i>)						
Base MLLM		49.3	60.5	36.9	85.3	18.1
AttWarp (<i>ours</i>)	Rectilinear warping	58.1	63.7	40.4	87.5	25.5
AttWarp-Chain (<i>ours</i>)	Adaptive Chains	60.3	64.4	41.6	88.2	27.6
Qwen (Yang et al., 2024a) (<i>Cross-attention VL adapter & partially closed data</i>)						
Base MLLM		81.0	62.4	47.3	86.1	77.3
AttWarp (<i>ours</i>)	Rectilinear warping	84.7	64.0	50.4	87.4	84.1
AttWarp-Chain (<i>ours</i>)	Adaptive Chains	85.9	64.8	51.0	88.0	85.3

AttWarp vs. other distorted or augmented inputs



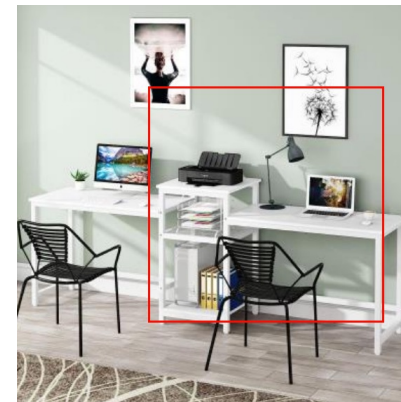
(Green Masking)



(Visual Grounding)



(Alpha Blending)



(Cropping)



(Warping)

Methods	Key Technique	TextVQA	GQA	MMMU	POPE	DocVQA
FGVP-mask (Yang et al., 2023b)	Green mask overlay	39.4	59.2	36.1	85.3	19.0
FGVP-blur (Yang et al., 2023b)	Blur background	33.9	59.5	35.0	83.1	18.6
SoM (Yang et al., 2023a)	Grounded segments	18.8	54.5	35.6	78.5	15.8
API (Yu et al., 2024)	Alpha channel fade	49.9	60.6	36.9	85.9	17.4
ViCrop (Zhang et al., 2025a)	Add object crop	56.3	60.9	37.2	87.0	22.5
AttWarp (<i>ours</i>)	Rectilinear warping	58.1	63.7	40.4	87.5	25.5
AttWarp-Chain (<i>ours</i>)	Adaptive Chains	60.3	64.4	41.6	88.2	27.6

Same results for detection

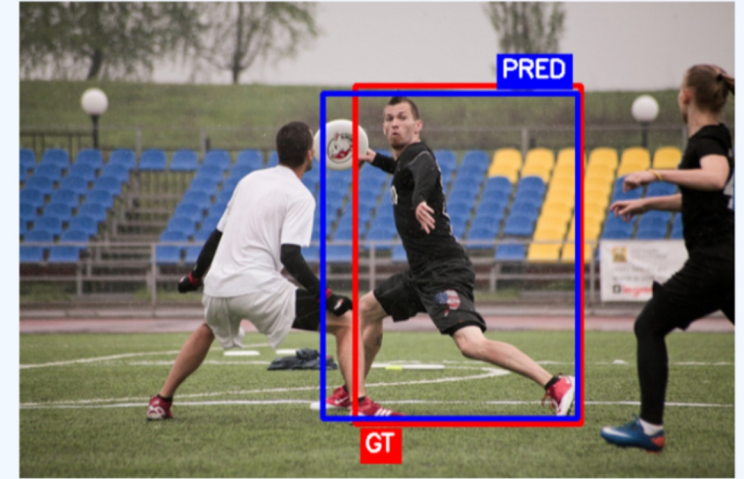
Referring expression: **Man in black facing us**



Original



Warped



Inversed Warped

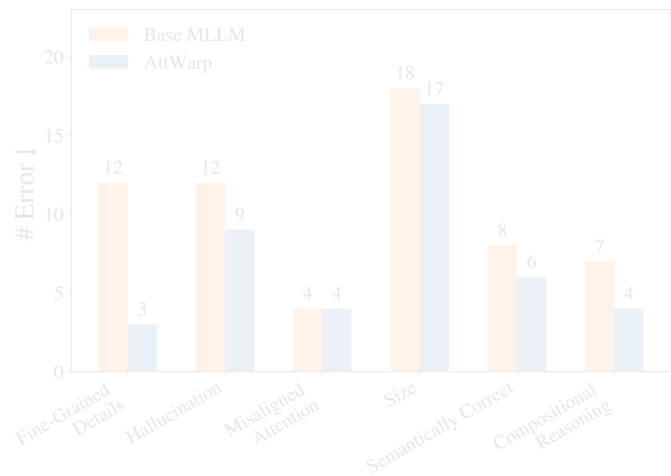
LISA-7B:
54% on RefCOCO

LISA-7B + AttWarp:
61% on RefCOCO

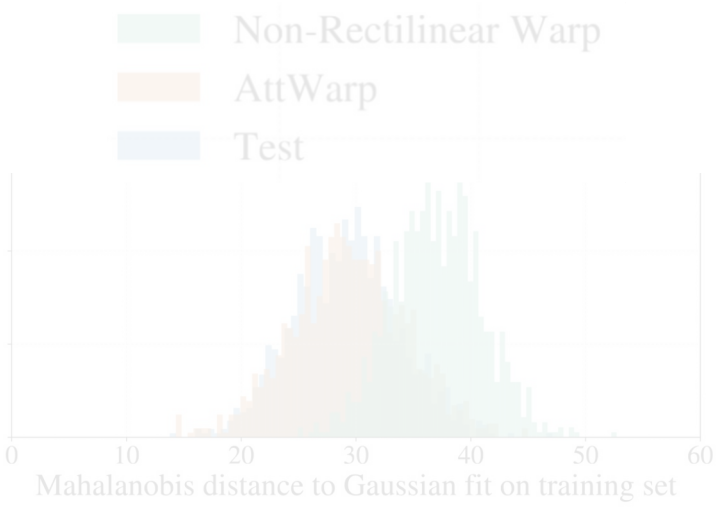
Insights



AttWarp improves attention



All failure modes go down



Rectilinear-ity is crucial

LLaVA		
Method	TextVQA	GQA
Base MLLM	49.3	60.5
AttWarp (Layer 20)	58.1	63.7
AttWarp (Layer 24)	58.4	62.8

Qwen		
Method	TextVQA	GQA
Base MLLM	81.0	62.4
AttWarp (Layer 16)	84.7	64.0
AttWarp (Layer 24)	84.2	63.6

Robust to design choices

AttWarp

- VLMs err even when they shouldn't
- Bridging cross-modal gap by warping images
- Allocate more/less pixels to relevant/irrelevant regions
- Tested, robust, intuitive, explainable